

VTOL Forward-Flight Aerodynamics: A Steady RANS Study of a Fixed-Wing VTOL UAV in Cruise

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Abstract

A full-airframe 3D external aerodynamics CFD simulation of a fixed-wing VTOL UAV in forward flight is presented, carried out in OpenFOAM with a steady $k-\omega$ SST RANS model on a 7 million cells mesh. The document goes over mesh generation and y^+ verification, surface-pressure and force integration, wing-section, wake and tip-vortex visualisation, and convergence assessment. At cruise ($U_\infty = 16$ m/s, $\rho = 1.2$ kg/m³, $Re_c \approx 1.4 \times 10^5$) and level attitude, the aircraft develops a lift coefficient $C_L = 0.558$ and a drag coefficient $C_D = 0.0623$, giving a lift-to-drag ratio of about 9.0 and 20.1 N of lift, enough to carry a 2 kg aircraft at 16 m/s.

Index Terms

CFD, RANS, OpenFOAM, VTOL UAV, external aerodynamics, $k-\omega$ SST, wing-tip vortices.

NOMENCLATURE

U_∞	freestream velocity, 16 m/s
ρ	air density, 1.2 kg/m ³
ν	kinematic viscosity, 1.5×10^{-5} m ² /s
q_∞	freestream dynamic pressure, Pa
b, c	wing span (1.8 m), chord (0.13 m)
S	planform reference area, 0.234 m ²
Re_c	chord Reynolds number
C_L, C_D, C_m	lift, drag, pitching-moment coefficients
C_p	pressure coefficient, $(p - p_\infty)/q_\infty$
L, D	lift and drag force, N
y^+	dimensionless wall distance
k, ω	turbulent kinetic energy; specific dissipation rate

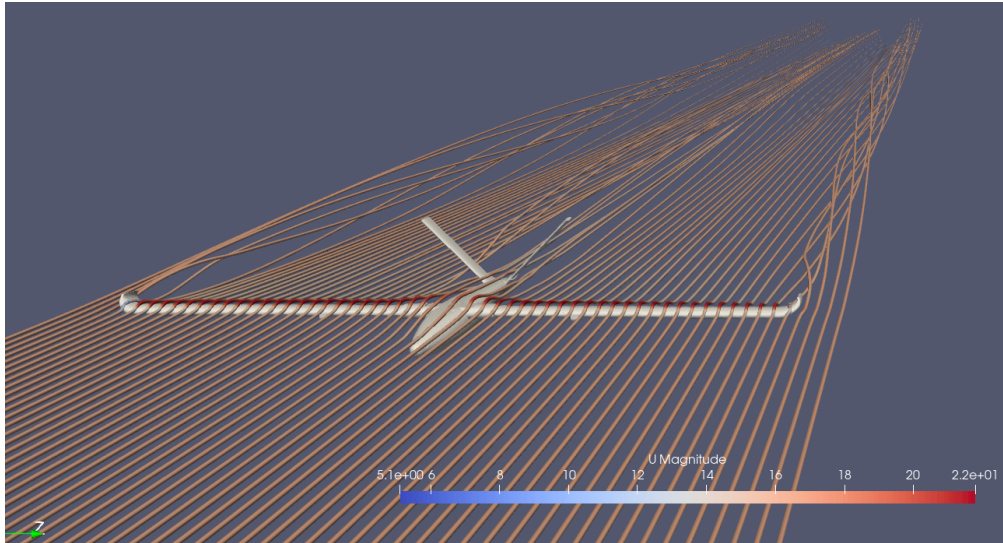


Fig. 1. Streamlines traced around the airframe in forward flight, coloured by velocity magnitude.

I. THE CASE

The aircraft in this simulation is a VTOL with a fixed, high-aspect-ratio wing (NACA 4421 section) with upturned winglets, a V-tail carried on twin booms, and dedicated lift-rotor arms. This study simulates the forward-

flight (cruise) condition at level attitude, in which the lift rotors are idle and the wing does the work, so the model is solved as a clean external-aerodynamics problem in 3D around the full airframe (Fig. 1). The design requirement is to carry 1.8–2.0 kg at about 16 m/s.

The freestream dynamic pressure and the chord Reynolds number follow from the cruise condition as

$$q_\infty = \frac{1}{2} \rho U_\infty^2 = 153.6 \text{ Pa}, \quad (1)$$

$$Re_c = \frac{U_\infty c}{\nu} \approx 1.4 \times 10^5, \quad (2)$$

placing the flow in the low-Reynolds regime typical of small UAVs. The full case definition is summarised in Table I.

TABLE I
CASE SUMMARY.

Solver	<code>foamRun</code> : incompressibleFluid, steady-state SIMPLEC (OpenFOAM 13)
Turbulence	RANS $k-\omega$ SST with wall functions (high- Re near-wall treatment)
Regime	incompressible cruise; $U_\infty = 16$ m/s; air $\rho = 1.2$ kg/m ³ ; level attitude
Reference	span $b = 1.8$ m; chord $c = 0.13$ m; planform area $S = 0.234$ m ² ; C_m about the wing quarter-chord
Reynolds	$Re_c \approx 1.4 \times 10^5$ ($\nu = 1.5 \times 10^{-5}$ m ² /s); fully turbulent model
Mesh	snappyHexMesh; hex-dominant; ≈ 7 M cells; no inflation layers
Compute	6-way MPI decomposition; 700 iterations; ≈ 1.7 h wall-clock

II. COMPUTATIONAL METHODOLOGY

A. Mesh and Near-Wall Resolution

A hex-dominant volume mesh wraps the airframe, with the surface triangulated finely over regions of high curvature (Figs. 2 and 3). There are no inflation layers: the near-wall region is captured by the snapped background cells and a graded refinement zone (Fig. 4), with wall functions bridging to the surface.

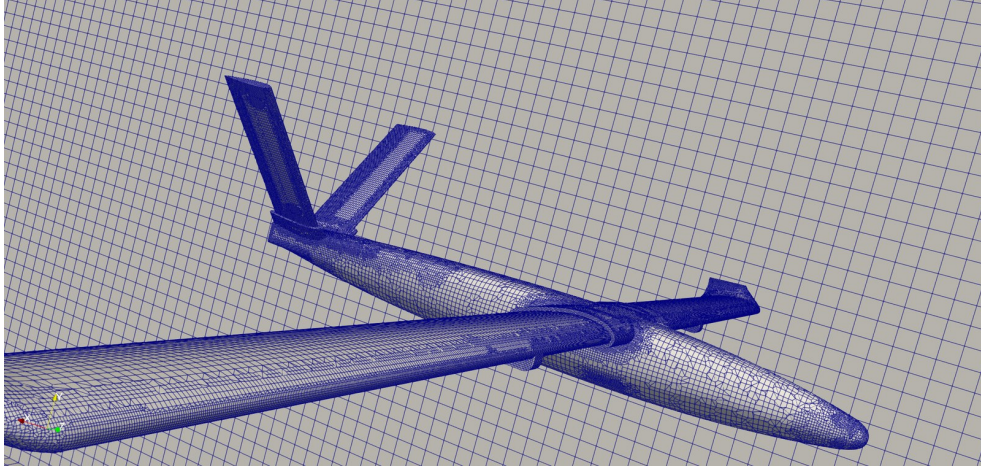


Fig. 2. Surface mesh over the full airframe.

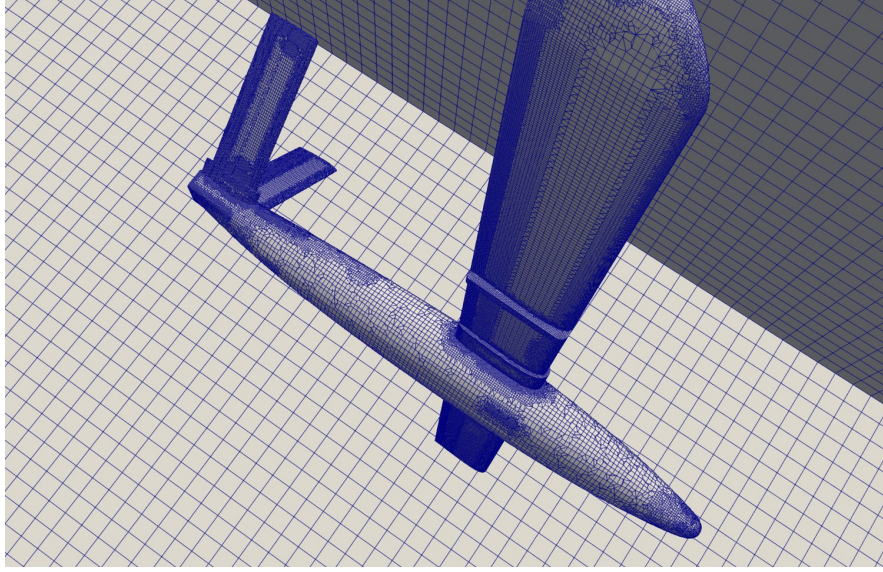


Fig. 3. Surface mesh: boom and fin detail.

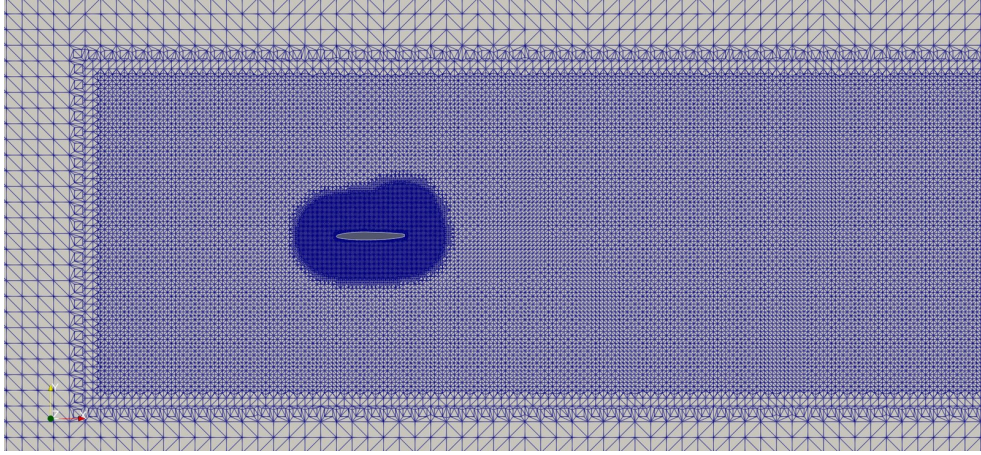


Fig. 4. Refinement regions: near-field region concentrates cells around the aircraft using snappyHexMesh distance method while the refinement box serves as an intermediary between the near-field and the coarser freestream.

B. Numerical Setup

Momentum is discretised with a second-order bounded upwind scheme for accuracy, the turbulence transport with first-order upwind for stability, and the pressure–velocity coupling uses SIMPLEC to permit aggressive under-relaxation. The inflow turbulence corresponds to a 5% intensity, $k = 1.5 (IU_\infty)^2 = 0.96 \text{ m}^2/\text{s}^2$, with a turbulent-to-laminar viscosity ratio of 100, $\omega = k/(100\nu) = 640 \text{ s}^{-1}$, representative of flight in disturbed low-altitude air. The governing dictionaries are listed in Table II.

TABLE II
BOUNDARY CONDITIONS AND SOLVER DICTIONARIES.

Boundary conditions	
Inlet	$U = (16\ 0\ 0)$ m/s fixed; k, ω fixed; p zeroGradient
Outlet	$p = 0$ fixedValue; U, k, ω inletOutlet
Far-field	slip on the top, bottom and side boundaries
Aircraft	no-slip; ν_t : Spalding wall function; ω : blended wall function; k : kqRWallFunction
Freestream	$k = 0.96\ \text{m}^2/\text{s}^2$; $\omega = 640\ \text{s}^{-1}$
Discretisation (fvSchemes)	
Time	steadyState
Gradient	Gauss linear; cell-limited on ∇U
Divergence	bounded linearUpwindV (U); bounded upwind (k, ω)
Laplacian	Gauss linear corrected
Coupling and solvers (fvSolution)	
Algorithm	SIMPLEC (consistent); 1 non-orthogonal corrector
Pressure	GAMG + Gauss-Seidel, tol 10^{-7} , relTol 0.01
$U/k/\omega$	smoothSolver (Gauss-Seidel), tol 10^{-8} , relTol 0.1
Relaxation	$U = 0.9$; $k, \omega = 0.5$

C. Wall Resolution (y^+)

With wall functions, the near-wall cells should sit in the log-law band, $y^+ \gtrsim 30$. Over the airframe the area-averaged value is $y^+ = 49.8$ (range 1.3–298). Mapped over the surface (Figs. 5 and 6), almost the entire wing and fuselage sit at $y^+ \geq 30$; values below 30 occur only in thin strips along the leading and trailing edges, on parts of the V-tail ($y^+ \approx 15$ –25), in the wing–fuselage junction corners, and at the rod mounts and wingtips. The wall-function assumption therefore holds over most of the wetted surface; in the remaining regions the Spalding (continuous) wall function for ν_t and the blended ω wall function cover the buffer layer.

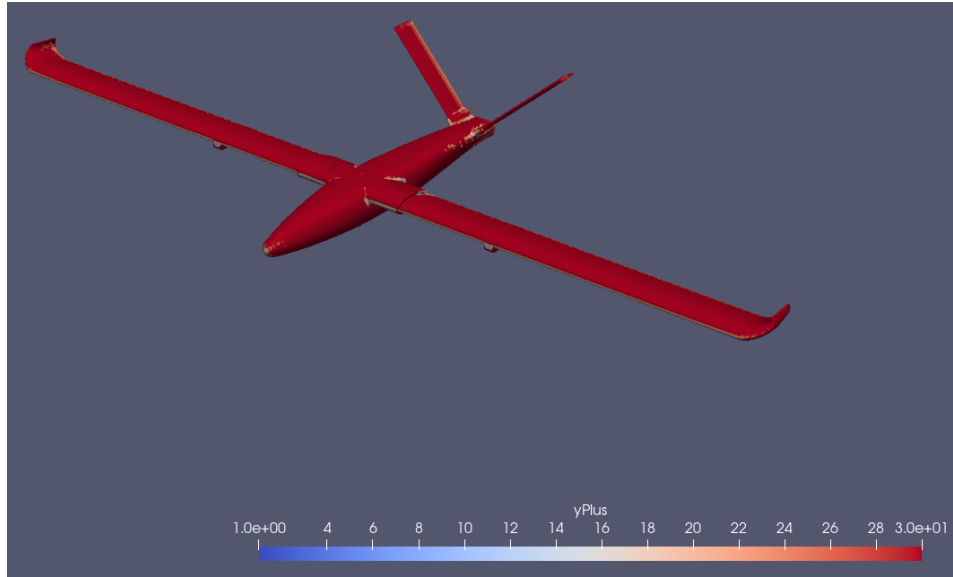


Fig. 5. y^+ clamped to 1–30: the surface is almost entirely ≥ 30 (red).

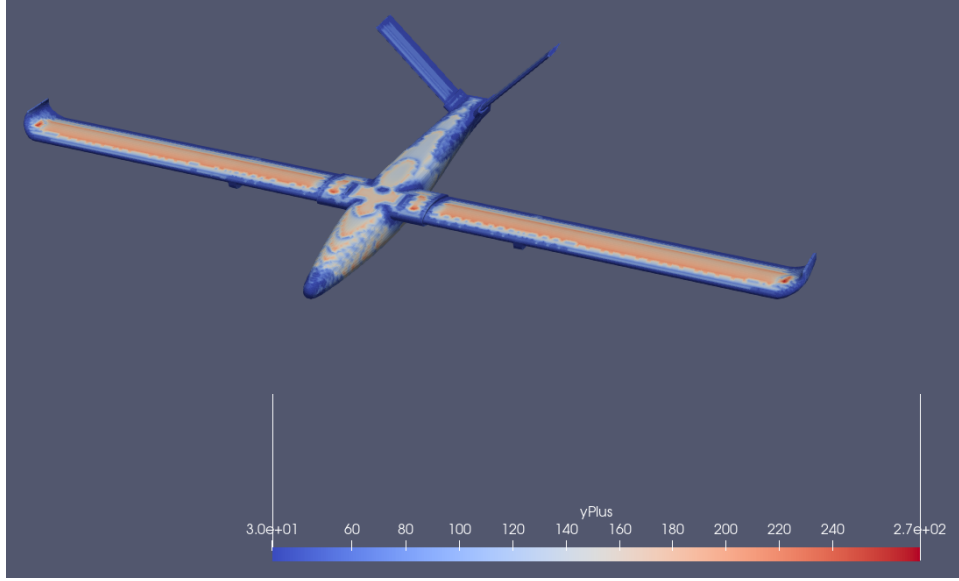


Fig. 6. y^+ over the airframe, shown from 30 to 270.

III. RESULTS

The converged aerodynamic coefficients, averaged over iterations 200–700, are collected in Table III; the surface-pressure field, the wing section and the wake are discussed in turn below.

TABLE III
KEY AERODYNAMIC RESULTS (LEVEL ATTITUDE, $U_\infty = 16$ m/s, MEAN OF ITERATIONS 200–700).

C_L	0.558	L	20.1 N	L/D	9.0
C_D	0.0623	D	2.24 N	C_m (c/4, + nose-down)	0.082

A. Surface Pressure and Loads

The surface pressure coefficient (Figs. 7–9) shows strong suction over the front third of the wing upper surface, weakening towards the trailing edge with no sign of separation, while the lower surface stays near $C_p \approx 0$. Stagnation occurs along the wing leading edge and at the fuselage nose. The force coefficients are defined as

$$C_L = \frac{L}{q_\infty S}, \quad C_D = \frac{D}{q_\infty S}, \quad (3)$$

and integrating the surface tractions yields $C_L = 0.558$ and $C_D = 0.0623$ (Table III). Re-dimensionalising through (3),

$$L = C_L q_\infty S \approx 20.1 \text{ N } (\approx 2.05 \text{ kgf}), \quad D \approx 2.24 \text{ N}, \quad (4)$$

so at level attitude the airframe supports a 2.0 kg aircraft at 16 m/s, at $L/D \approx 9.0$. Equivalently, level cruise is reached at 15.0 m/s for 1.8 kg and 15.8 m/s for 2.0 kg. The lift coefficient is consistent with a lifting-line estimate from 2D section data ($C_L \approx 0.6$). Small local pressure irregularities at the wing–fuselage junction and the rod mounts coincide with the low- y^+ regions of Fig. 5.

The pitching moment about the wing quarter-chord is $C_m = 0.082$ (nose-down), but this alone does not give the static stability of the aircraft; the trim condition depends on the center-of-gravity position and the tail setting.

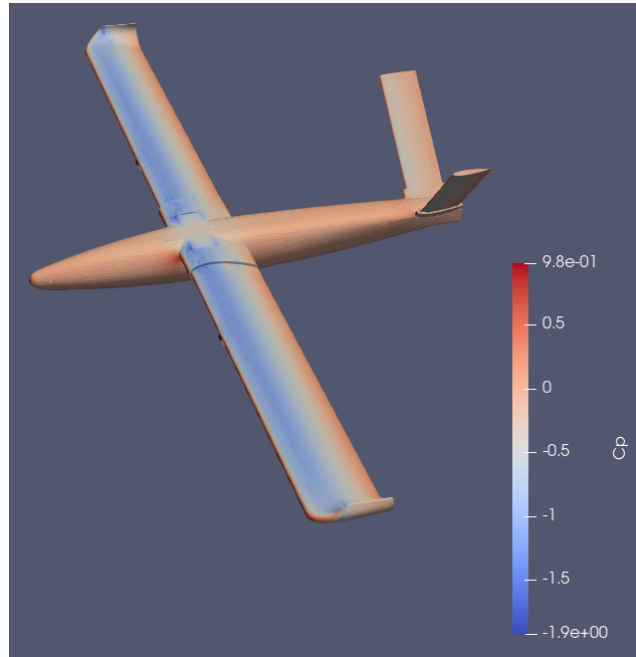


Fig. 7. Pressure coefficient on the upper surface: suction over the front of the wing.

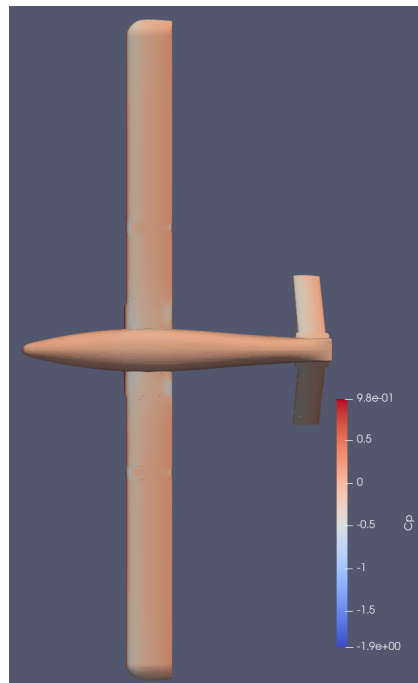


Fig. 8. Pressure coefficient on the lower surface ($C_p \approx 0$ to slightly positive).

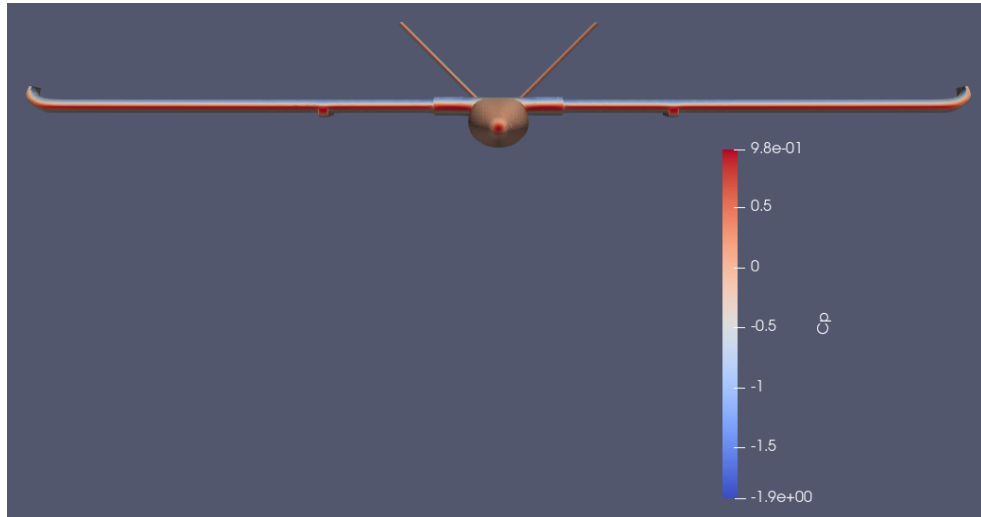


Fig. 9. Pressure coefficient, head-on: stagnation along the wing leading edge and at the nose.

B. Wing Section

A cut through the wing (Figs. 10 and 11) shows the stagnation point on the leading edge, the suction peak on the forward upper surface and attached flow to the trailing edge, with a thin wake. The flow accelerates to about 22 m/s over the upper surface, with Bernoulli's equation giving $C_p = 1 - (22/16)^2 \approx -0.9$, which is consistent with the suction seen in the pressure field. Without inflation layers, the snapped near-wall cells appear as small steps in the section outline near the trailing edge.

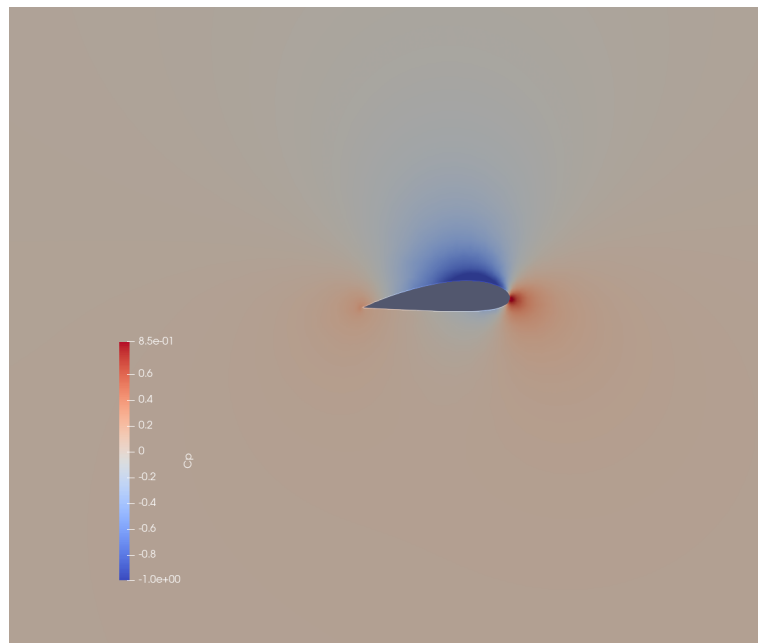


Fig. 10. Pressure coefficient on a wing section (flow from right to left).

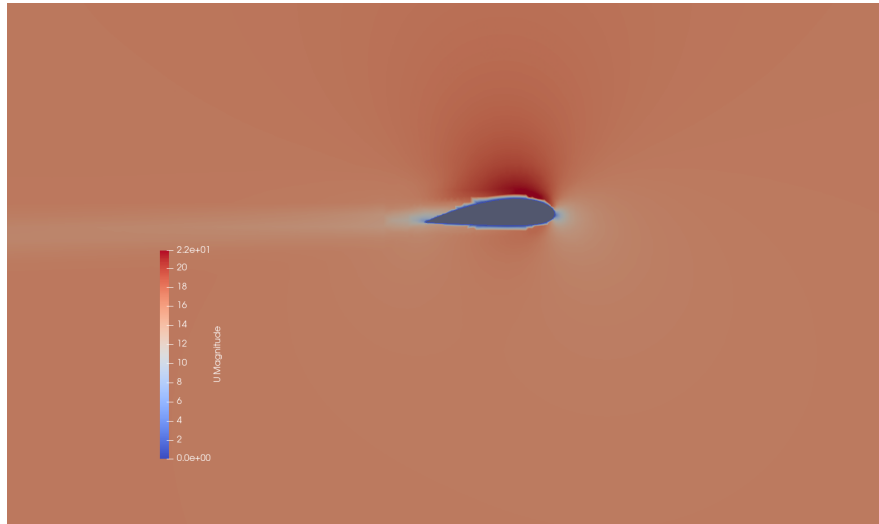


Fig. 11. Velocity magnitude on the same section (flow from right to left).

C. Wake and Wing-Tip Vortices

Streamlines seeded ahead of the aircraft and traced downstream reveal the wake structure (Figs. 1 and 12). At the wing tips the pressure difference drives air from the lower surface around the winglet onto the upper surface, rolling the flow up into a coherent tip vortex, which is the signature of finite-wing induced drag (Figs. 13 and 14).

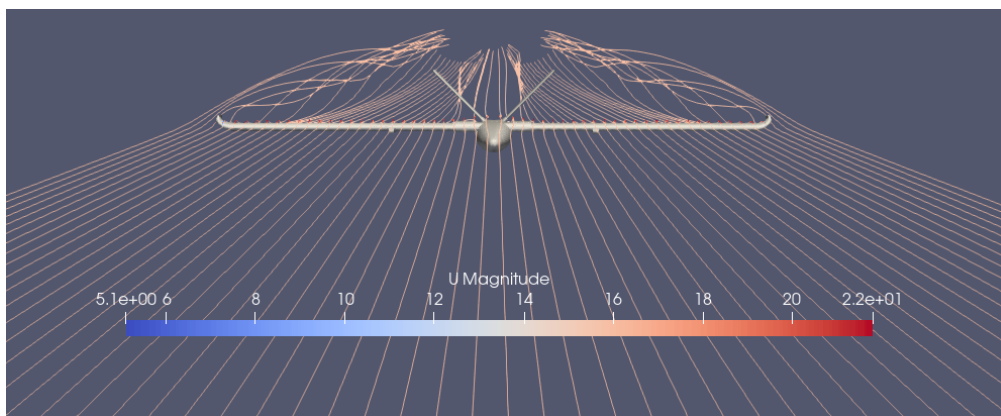


Fig. 12. Streamlines seen from the front: roll-up above the outboard wing.

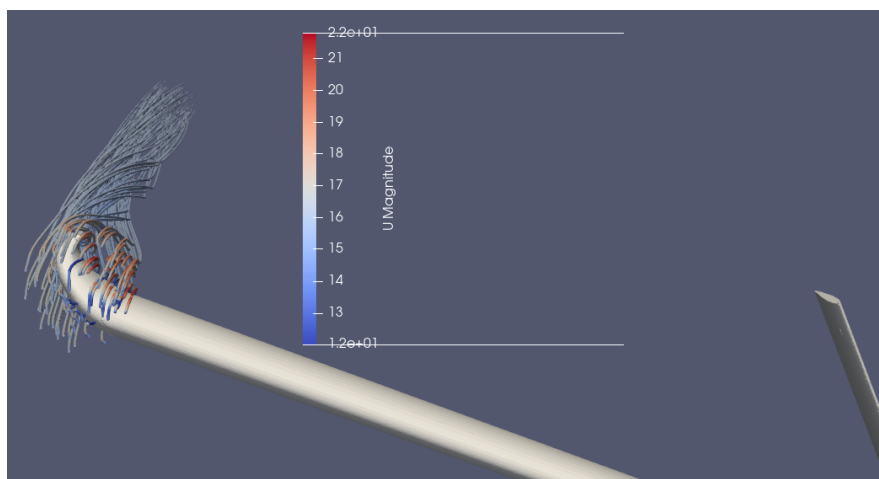


Fig. 13. Tip-vortex roll-up off the winglet, traced as streamtubes coloured by velocity magnitude.

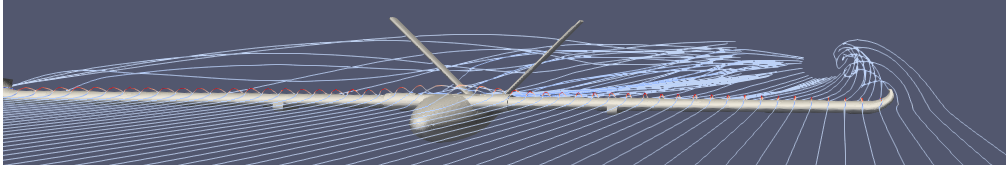


Fig. 14. Streamlines over the wing and the tip vortex rolling up behind the winglet.

IV. CONVERGENCE

The residual and force histories, extracted from the solver log, are shown in Figs. 15 and 16. The final residuals of momentum and pressure level off at 10^{-5} – 2×10^{-4} , and those of k and ω fall to 10^{-7} – 10^{-8} . The force coefficients settle by iteration 100 and then oscillate steadily without drift: C_L by about $\pm 3\%$ and C_D by about $\pm 2\%$ around the reported values (averaged from iteration 200 to 700). The solution is therefore converged in the averaged sense; the oscillation is typical of steady RANS around an aircraft.

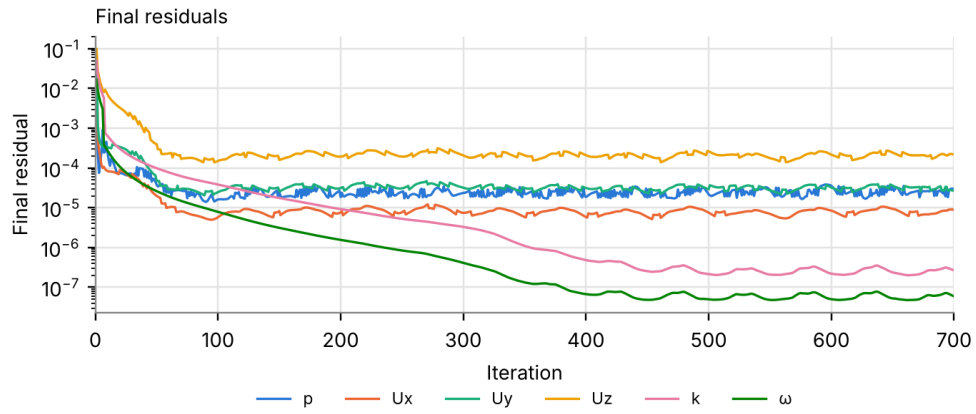


Fig. 15. Final residual history (log scale): p , U_x , U_y , U_z , k , ω .

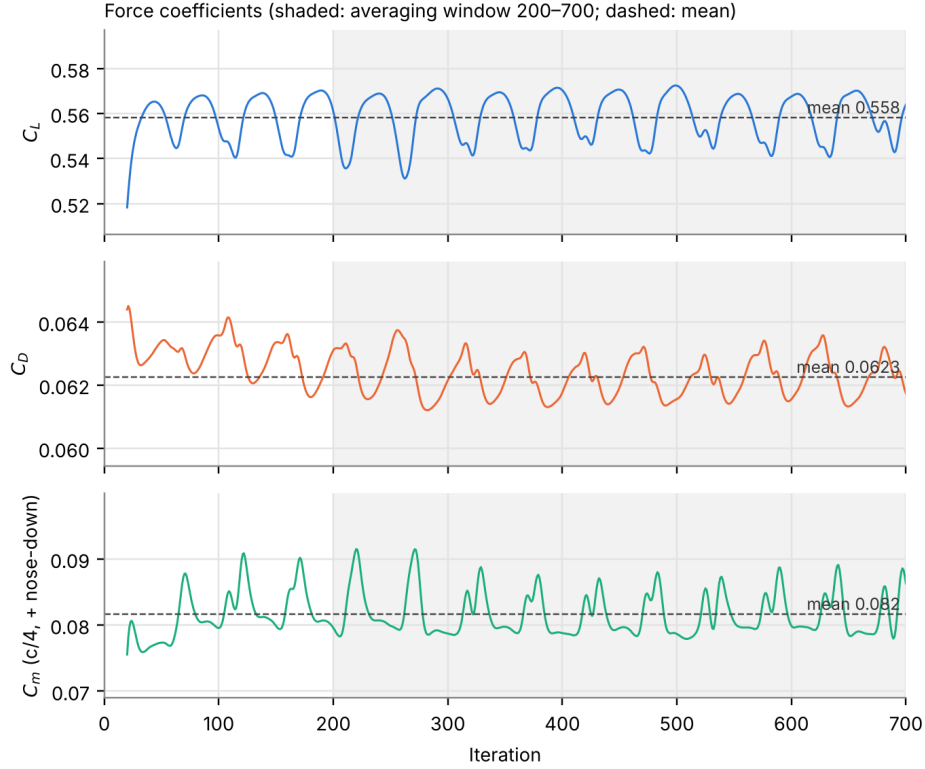


Fig. 16. Force-coefficient histories (C_L , C_D , C_m), shown from iteration 20; shaded: averaging window.

V. CONCLUSION

A converged, full-airframe RANS solution with wall-function boundary-layer treatment (y^+ mostly in the log-law band) of the VTOL UAV in level cruise at 16 m/s gives $C_L = 0.558$, $C_D = 0.0623$ and $L/D \approx 9.0$, enough lift for a 2 kg aircraft, with physically consistent surface loads, attached flow over the wing and clean wing-tip vortex roll-up. The workflow: meshing, solver setup, y^+ and convergence verification, and post-processing, is presented end to end. This study is only to be used as an educated engineering estimate to design our scaled-down VTOL prototype since no mesh independence analysis with GCI and validation was done, the mesh used wall functions without prism layers to model the boundary layer and a steady, fully turbulent model with no laminar-turbulent transition modeling was used on a low-Re, mildly oscillating flow.